Kubernetes-02

1.Created a Simple Nginx Pod Using YAML

- Created a YAML configuration file to define a Kubernetes Pod.
- Configured an Nginx container using the latest image.
- Deployed the Pod using `kubectl apply`.
- Verified Pod status using `kubectl get pods`.
- Inspected Pod configuration using `kubectl describe`.
- Checked container logs using `kubectl logs`.

```
apiVersion: v1
kind: Pod
metadata:
  name: firstpod
spec:
  containers:
    - name: nginx
      image: nginx:latest
      ports:
        - containerPort: 80
```

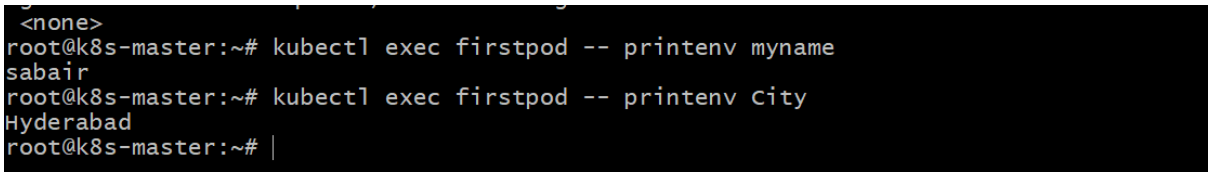
```
root@k8s-master:~# vi firstpod.yml
root@k8s-master:~# kubectl apply -f firstpod.yml
pod/firstpod created
root@k8s-master:~# kubectl get pods
NAME                        READY   STATUS    RESTARTS   AGE
firstpod                    1/1     Running   0          55s
```

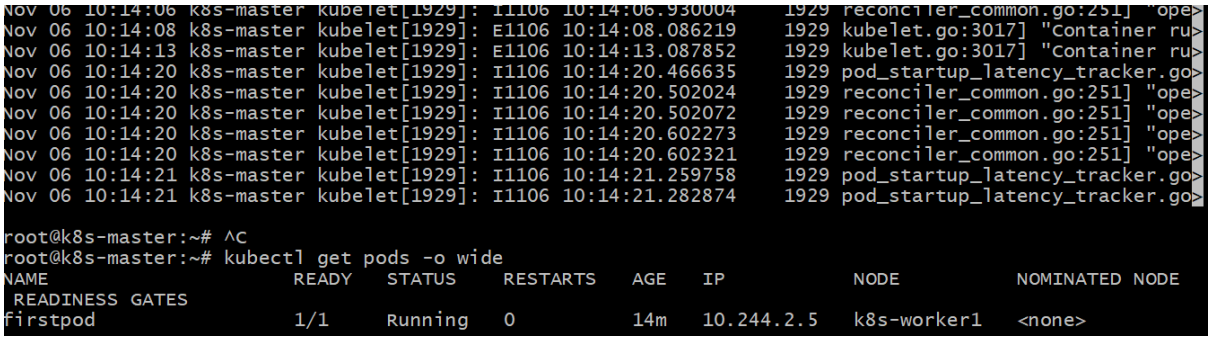
```
root@k8s-master:~# kubectl get pod firstpod -o wide
NAME       READY   STATUS    RESTARTS   AGE     IP           NODE          NOMINATED NODE   READINESS G
ATES
firstpod   1/1     Running   0          3m25s   10.244.2.4   k8s-worker1   <none>           <none>
root@k8s-master:~#
```

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2.Configuring Environment Variables in a Kubernetes Pod

1. **Modified the Pod YAML file to configure environment variables.**
2. **Added `myname` with the value `sabair`.**
3. **Added `City` with the value `Hyderabad`.**
4. **Deleted and recreated the existing Pod to apply the updated configuration.**
5. **Verified the environment variables using `kubectl exec`.**
6. **Successfully confirmed both environment variable values inside the Nginx container.**





3.Deploy a Pod with Commands (Args) in YAML

We are using `command` to override the container's default startup command and `args` to pass `sleep 50`.

```
Name:             firstpod
Namespace:        default
Priority:         0
Service Account:  default
Node:             k8s-worker1/172.31.30.88
Start Time:       Fri, 06 Nov 2026 13:53:02 +0000
Labels:           <none>
Annotations:      <none>
Status:           Running
IP:               10.244.2.6
IPs:
  IP:  10.244.2.6
Containers:
  nginx:
    Container ID:  containerd://7a51ec9800354deb58046a2b285d5d3825dd4d780fa9b79489b47d7eb444f0ca
    Image:         nginx:latest
    Image ID:      docker.io/library/nginx@sha256:165ddde8751e75a1c907d24ef10d9f6deb08b57e0fe56caf67512
486aff76d82
    Port:          80/TCP
    Host Port:     0/TCP
    Command:
      /bin/sh
      -c
    Args:
      sleep 50
    State:          Running
      Started:      Fri, 06 Nov 2026 13:53:04 +0000
    Ready:          True
    Restart Count:  0
```

Delete the existing Pod

kubectl delete pod firstpod

Deploy the updated YAML

kubectl apply -f firstpod.yml

Verify Pod status

kubectl get pods

```
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-lqhhh (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-lqhhh:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    Optional:                false
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason     Age   From               Message
  ----    ------     ----  ----               -------
  Normal  Scheduled  25s   default-scheduler  Successfully assigned default/firstpod to k8s-worker1
  Normal  Pulling    25s   kubelet            Pulling image "nginx:latest"
  Normal  Pulled     23s   kubelet            Successfully pulled image "nginx:latest" in 1.245s (1.245
s including waiting). Image size: 63933865 bytes.
  Normal  Created    23s   kubelet            Created container: nginx
  Normal  Started    23s   kubelet            Started container nginx
root@k8s-master:~#
```

```
    Image ID:      docker.io/library/nginx@sha256:165ddde8751e75a1c907d24ef10d9f6deb08b57
486aff76d82
    Port:          80/TCP
    Host Port:     0/TCP
    Command:
      /bin/sh
      -c
    Args:
      sleep 50
    State:          Running
      Started:      Fri, 06 Nov 2026 13:53:04 +0000
    Ready:          True
    Restart Count:  0
    Environment:
      myname:  sabair
      City:    Hyderabad
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-lqhhh (ro)
Conditions:
  Type                        Status
```

```
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-lqhhh:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    Optional:                false
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason     Age   From               Message
  ----    ------     ----  ----               -------
  Normal  Scheduled  25s   default-scheduler  Successfully assigned default/firstpod to k8s-worker1
  Normal  Pulling    25s   kubelet            Pulling image "nginx:latest"
  Normal  Pulled     23s   kubelet            Successfully pulled image "nginx:latest" in 1.245s (1.245
s including waiting). Image size: 63933865 bytes.
  Normal  Created    23s   kubelet            Created container: nginx
  Normal  Started    23s   kubelet            Started container nginx
root@k8s-master:~#
```

4.Create a Pod with Two Containers

Objective: Create a single Kubernetes Pod containing two Nginx containers and verify that both containers can communicate through the same Pod network.

- Created a Pod containing two Nginx containers.
- Configured the first Nginx container to listen on port 80.
- Configured the second Nginx container to listen on port 8081.
- Used separate ports because containers in the same Pod share a network namespace.
- Verified communication between both containers.

apiVersion: v1

```
kind: Pod
metadata:
  name: firstpod
spec:
  containers:
    - name: nginx1
      image: nginx:latest
      ports:
        - containerPort: 80


    - name: nginx2
      image: nginx:latest
      command: ["/bin/sh", "-c"]
      args:
        - |
          echo 'server {
            listen 8081;
            location / {
              root /usr/share/nginx/html;
              index index.html;
            }
          }' > /etc/nginx/conf.d/default.conf
          exec nginx -g 'daemon off;'
      ports:
        - containerPort: 8081
```

```
root@k8s-master:~# kubectl delete pod firstpod
pod "firstpod" deleted from default namespace
root@k8s-master:~# kubectl apply -f firstpod.yml
pod/firstpod created
root@k8s-master:~# kubectl get pods
NAME                    READY   STATUS    RESTARTS      AGE
firstpod                1/2     Error     1 (13s ago)   19s
nginx-66686b6766-dj5nf  1/1     Running   0             152m
nginx-66686b6766-ffc2q  1/1     Running   0             3h5m
nginx-66686b6766-ms8qn  1/1     Running   0             152m
root@k8s-master:~# kubectl exec firstpod -c nginx2 -- grep listen /etc/nginx/conf.d/default.conf
error: Internal error occurred: unable to upgrade connection: container not found ("nginx2")
root@k8s-master:~# vi firstpod.yml
root@k8s-master:~# kubectl delete pod firstpod
pod "firstpod" deleted from default namespace
root@k8s-master:~# kubectl apply -f firstpod.yml
pod/firstpod created
root@k8s-master:~# kubectl get pods
NAME                    READY   STATUS    RESTARTS   AGE
firstpod                2/2     Running   0          7s
nginx-66686b6766-dj5nf  1/1     Running   0          155m
nginx-66686b6766-ffc2q  1/1     Running   0          3h7m
nginx-66686b6766-ms8qn  1/1     Running   0          155m
root@k8s-master:~# kubectl exec firstpod -c nginx1 -- curl http://localhost:8081
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
```

```
root@k8s-master:~# kubectl exec firstpod -c nginx1 -- curl http://localhost:8081
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100   896  100   896    0     0   625k      0 --:--:-- --:--:-- --:--:--  875k
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>
```

5.Configuring an Init Container in Kubernetes

1. Added an Init Container to the Pod configuration.
2. Used the BusyBox image for the Init Container.
3. Configured the Init Container to sleep for 30 seconds.
4. Verified Init Container execution using Pod status and logs.
5. Confirmed that both Nginx containers started after Init Container completion.

```
root@k8s-master:~# vi firstpod.yml
root@k8s-master:~# kubectl delete pod firstpod
pod "firstpod" deleted from default namespace
root@k8s-master:~# kubectl apply -f firstpod.yml
pod/firstpod created
root@k8s-master:~# kubectl get pods
NAME                    READY   STATUS     RESTARTS        AGE
firstpod                0/2     Init:0/1   0               10s
nginx-66686b6766-dj5nf  1/1     Running    1 (8m48s ago)   19h
nginx-66686b6766-ffc2q  1/1     Running    1 (8m41s ago)   19h
nginx-66686b6766-ms8qn  1/1     Running    1 (8m48s ago)   19h
root@k8s-master:~# kubectl logs firstpod -c init-container
Init container started
root@k8s-master:~# kubectl get pods
NAME                    READY   STATUS    RESTARTS        AGE
firstpod                2/2     Running   0               51s
nginx-66686b6766-dj5nf  1/1     Running   1 (9m29s ago)   19h
nginx-66686b6766-ffc2q  1/1     Running   1 (9m22s ago)   19h
nginx-66686b6766-ms8qn  1/1     Running   1 (9m29s ago)   19h
root@k8s-master:~#
```

6.Generated Pod YAML Using kubectl

Objective: Generate a Kubernetes Pod YAML file using the `kubectl run` command, customize it, and deploy the Pod.

apiVersion: v1

kind: Pod

metadata:

  name: nginx

spec:

  containers:

    - name: nginx

      image: nginx

      ports:

        - containerPort: 80

```
status: {}
root@k8s-master:~# vi nginx-pod.yml
root@k8s-master:~# kubectl apply -f nginx-pod.yml
Error from server (BadRequest): error when creating "nginx-pod.yml": Pod in version "v1" cannot be handled as a Pod: stric
t decoding error: unknown field "spec.containers[1].containerPort"
root@k8s-master:~# vi nginx-pod.yml
root@k8s-master:~# kubectl apply -f nginx-pod.yml
pod/nginx created
root@k8s-master:~# kubectl get pods
NAME                    READY   STATUS    RESTARTS      AGE
firstpod                2/2     Running   0             14m
nginx                   1/1     Running   0             8s
```

7.Difference Between kubectl create and kubectl apply

Comparing kubectl create and kubectl apply

1. Deleted the previously existing Nginx Pod.
2. Created a new Pod using `kubectl create -f`.
3. Verified the Pod status using `kubectl get pods`.
4. Attempted to create the same Pod again and observed the `AlreadyExists` error.
5. Used `kubectl apply -f` to update the existing Pod configuration.
6. Observed the missing last-applied-configuration annotation warning.

```
root@k8s-master:~# vi nginx-pod.yml
root@k8s-master:~# kubectl create -f nginx-pod.yml
Error from server (AlreadyExists): error when creating "nginx-pod.yml": pods "nginx" already exists
root@k8s-master:~# delete pod nginx
delete: command not found
root@k8s-master:~# kubectl delete pod nginx
pod "nginx" deleted from default namespace
root@k8s-master:~# vi nginx-pod.yml
root@k8s-master:~# kubectl create -f nginx-pod.yml
pod/nginx created
root@k8s-master:~# kubectl get pods
NAME                     READY   STATUS    RESTARTS      AGE
firstpod                 2/2     Running   0             24m
nginx                    1/1     Running   0             11s
nginx-66686b6766-dj5nf   1/1     Running   1 (32m ago)   19h
nginx-66686b6766-ffc2q   1/1     Running   1 (32m ago)   20h
nginx-66686b6766-ms8qn   1/1     Running   1 (32m ago)   19h
root@k8s-master:~# kubectl create -f nginx-pod.yml
Error from server (AlreadyExists): error when creating "nginx-pod.yml": pods "nginx" already exists
root@k8s-master:~# kubectl apply -f nginx-pod.yml
Warning: resource pods/nginx is missing the kubectl.kubernetes.io/last-applied-configuration annotation which is required
by kubectl apply. kubectl apply should only be used on resources created declaratively by either kubectl create --save-con
fig or kubectl apply. The missing annotation will be patched automatically.
pod/nginx configured
root@k8s-master:~#
```

8.Edit a Pod Using kubectl edit

Objective: Modify an existing Pod configuration using `kubectl edit` and understand which Pod fields are immutable.

- Used `kubectl edit pod nginx` to modify the Pod.
- Attempted to update the container image.
- Investigated the validation error using the temporary YAML file.
- Updated the container image to `nginx:latest`.
- Verified the container restart through Kubernetes events.

- Confirmed that the updated container started successfully.

```
                        node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason     Age                From               Message
  ----    ------     ----               ----               -------
  Normal  Scheduled  15m                default-scheduler  Successfully assigned default/nginx to k8s-worker2
  Normal  Pulling    15m                kubelet            Pulling image "nginx"
  Normal  Pulled     14m                kubelet            Successfully pulled image "nginx" in 1.248s (1.249s including w
aiting). Image size: 63933865 bytes.
  Normal  Killing    28s                kubelet            Container nginx definition changed, will be restarted
  Normal  Pulling    28s                kubelet            Pulling image "nginx:latest"
  Normal  Created    26s (x2 over 14m)  kubelet            Created container: nginx
  Normal  Started    26s (x2 over 14m)  kubelet            Started container nginx
  Normal  Pulled     26s                kubelet            Successfully pulled image "nginx:latest" in 1.246s (1.246s incl
uding waiting). Image size: 63933865 bytes.
```

9.Exposing a Pod Using a ClusterIP Service

- Identified that the Pod had no labels.
- Added the label `app=nginx` to the Pod.
- Created a ClusterIP Service named `firstpod-service`.
- Configured Service port 80 and target port 80.
- Verified the Service and its Pod endpoint.

```
root@k8s-master:~# kubectl expose pod firstpod --name=firstpod-service --port=80 --target-port=80 --type=ClusterIP
error: couldn't retrieve selectors via --selector flag or introspection: the pod has no labels and cannot be exposed
root@k8s-master:~#
root@k8s-master:~# kubectl get pod firstpod --show-labels
NAME       READY   STATUS    RESTARTS   AGE   LABELS
firstpod   2/2     Running   0          45m   <none>
root@k8s-master:~# kubectl label pod firstpod app=nginx
pod/firstpod labeled
root@k8s-master:~# kubectl expose pod firstpod --name=firstpod-service --port=80 --target-port=80 --type=ClusterIP
service/firstpod-service exposed
root@k8s-master:~# kubectl get svc firstpod-service
NAME               TYPE        CLUSTER-IP    EXTERNAL-IP   PORT(S)   AGE
firstpod-service   ClusterIP   10.97.27.86   <none>        80/TCP    15s
root@k8s-master:~# kubectl describe svc firstpod-service
Name:                     firstpod-service
Namespace:                default
Labels:                   app=nginx
Annotations:              <none>
Selector:                 app=nginx
Type:                     ClusterIP
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.97.27.86
IPs:                      10.97.27.86
Port:                     <unset>  80/TCP
TargetPort:               80/TCP
Endpoints:                10.244.1.4:80,10.244.2.13:80,10.244.1.5:80 + 1 more...
Session Affinity:         None
Internal Traffic Policy:  Cluster
Events:                   <none>
root@k8s-master:~#
```

10.Configuring CPU and Memory Requests and Limits in Kubernetes

- Added CPU and memory requests to both Nginx containers.
- Configured CPU limits of 200m for each container.
- Configured memory limits of 128Mi for each container.
- Deleted and recreated the Pod with updated resource configuration.
- Verified resource requests and limits using `kubectl describe pod`.

```
root@k8s-master:~# kubectl get pods
NAME                        READY   STATUS    RESTARTS      AGE
firstpod                    2/2     Running   0             68s
```

```
Events:
  Type    Reason     Age   From               Message
  ----    ------     ----  ----               -------
  Normal  Scheduled  100s  default-scheduler  Successfully assigned default/firstpod to k8s-worker1
  Normal  Pulling    101s  kubelet            Pulling image "busybox"
  Normal  Pulled     100s  kubelet            Successfully pulled image "busybox" in 1.219s (1.219s including waiting). Im
age size: 2236767 bytes.
  Normal  Created    100s  kubelet            Created container: init-container
  Normal  Started    100s  kubelet            Started container init-container
  Normal  Pulling    69s   kubelet            Pulling image "nginx:latest"
  Normal  Pulled     68s   kubelet            Successfully pulled image "nginx:latest" in 1.239s (1.239s including waiting
). Image size: 63933865 bytes.
  Normal  Created    68s   kubelet            Created container: nginx1
  Normal  Started    68s   kubelet            Started container nginx1
  Normal  Pulling    68s   kubelet            Pulling image "nginx:latest"
  Normal  Pulled     66s   kubelet            Successfully pulled image "nginx:latest" in 1.258s (1.258s including waiting
). Image size: 63933865 bytes.
  Normal  Created    66s   kubelet            Created container: nginx2
  Normal  Started    66s   kubelet            Started container nginx2
root@k8s-master:~#
```

`kubectl get services`

```
root@k8s-master:~# kubectl get services
NAME              TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
firstpod-service  ClusterIP   10.97.27.86      <none>        80/TCP         11m
kubernetes        ClusterIP   10.96.0.1        <none>        443/TCP        21h
nginx             NodePort    10.101.139.199   <none>        80:32491/TCP   20h
```